

power source for its operation. For larger amplification, the voltage amplifier is formed by cascading a number of voltage doublers in series (Fig. 3).

A DC converter—a rectifier for RF signals—converts the received small voltage of RF signals and stores harvested energy in a capacitor. The harvester also has a power management system that takes care of utilisation of stored power, which can later be utilised for different applications.

Ambient backscatter transmitter. Backscattering of RF signals is based on the basic principle of reflection and absorption of electromagnetic waves. When an electromagnetic wave encounters a boundary between two transmission media that have different impedances/densities, it is reflected or absorbed. Thus, backscattering of RF signals is achieved by modulating the impedance of an antenna in the presence of an incident signal.

The backscatter transmitter includes a switch (load modulator) that modulates impedance of the antenna and causes a change in the amount of energy reflected by the antenna (Fig. 4). Input signals of the switch are a sequence of bits 1 and 0, which the switch uses to toggle the antenna between backscatter (reflective) and non-backscatter (absorptive) states to send bits to the receiver. Reflective state represents bit 1, and absorptive state represents bit 0.

Ambient backscatter receiver. To demodulate from backscatter signals, the receiver adopts a simple circuit composed of three main components: envelope averager, threshold calculator and comparator (Fig. 5). Received signal is first smoothened to average out the variations due to modulation.

Next, the threshold calculator computes a threshold by taking the

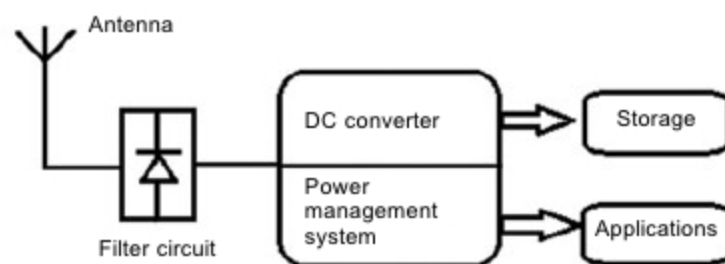


Fig. 3: RF energy harvesting circuit

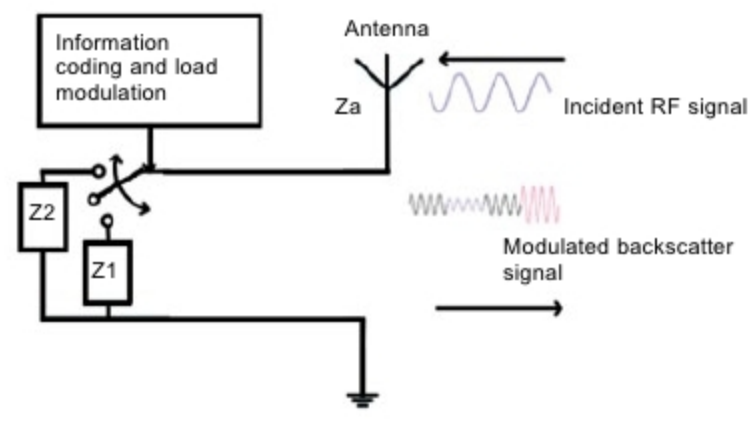


Fig. 4: Transmitter design

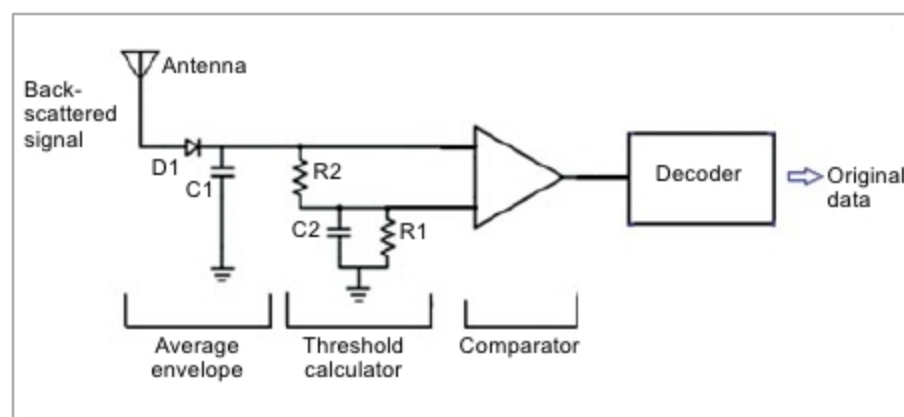


Fig. 5: Circuit diagram for ambient backscatter receiver

mean of the two voltage levels.

Then, by comparing instantaneously generated voltage at the first step with the threshold, it interprets the received signals into a stream of information bits.

The ambient backscatter receiver is implemented using an envelope detector and resistive/capacitive (RC) circuit to smoothen out natural variations in RF signals (TV signals). Output of the averaging circuit produces two signal levels (say, V_1 and V_0 ; $V_1 > V_0$), corresponding to bits 1 and 0.

To distinguish between the two, the receiver first computes a threshold value using threshold computation circuitry. The produced signal level along with threshold value is given to the comparator, which then produces bits 1 and 0 corresponding to V_1 and V_0 . Output of the comparator is then passed to the micro-

controller for decoding and further processing. Averaging mechanism is used for decoding the received signal.

Designing ambient backscatter

Designing of ambient backscatter is challenging due to the following reasons.

First, it uses uncontrollable RF signals that already have information encoded in these. Hence, it requires a different mechanism to extract backscattered information.

Second, a general backscatter receiver uses power consuming components such as oscillators and analogue-to-digital converters (ADCs) and decodes the signals with relatively complex digital signal processing techniques. These techniques are not practical for use in a battery-free receiver.

Last, ambient backscatter does not have a centralised controller to coordinate all communication. For this, the designer needs to develop its own protocol for information transmission and reception.

Advantages. There are several advantages of ambient backscatter.

1. It is battery-free. While it can only collect microwatts of electricity, it is enough to power itself and small onboard sensors such as LEDs, touch sensors, low power displays and the like.

2. It does not need any special purpose chips and other analogue components. It can be manufactured using commonly available components.

3. Its size is relatively small—about the size of a carrom coin.

4. Cost of manufacturing is quite low—about ₹ 50 each.

5. It does not need a dedicated power source (unlike in RFID communication), since it leverages radio signals that are already around.

Limitations. Ambient backscatter technology has certain limitations.

1. As backscatter transmitters